Experiment – 05

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Branch: BE-CSE(AIML)

Subject: Advanced Database Management System

Question 1:

Julia asked her students to create some coding challenges. Write a query to print the hacker_id, name, and the total number of challenges created by each student. Sort your results by the total number of challenges in descending order. If more than one student created the same number of challenges, then sort the result by hacker_id. If more than one student created the same number of challenges and the count is less than the maximum number of challenges created, then exclude those students from the result.

Input Format

The following tables contain challenge data:

Column	Type
hacker_id	Integer
name	String

- Hackers: The hacker_id is the id of the hacker, and name is the name of the hacker.
- Challenges: The challenge_id is the id of the challenge, and hacker_id is the id of the student who created the challenge.

Column	Type
challenge_id	Integer
hacker_id	Integer

Solution:

```
SELECT
    H.HACKER_ID,
    H.NAME,
    COUNT(C.CHALLENGE_ID) AS TOTAL_CHALLENGES
FROM HACKERS AS H
JOIN CHALLENGES AS C
    ON H.HACKER_ID = C.HACKER_ID
GROUP BY
    H.HACKER_ID,
    H.NAME
HAVING
```

```
  COUNT(C.CHALLENGE_ID) = (
    SELECT MAX(CNT)
    FROM (
      SELECT COUNT(*) AS CNT
      FROM CHALLENGES
      GROUP BY HACKER_ID
    ) AS TEMP
  )
  OR COUNT(C.CHALLENGE_ID) IN (
    SELECT CNT
    FROM (
      SELECT COUNT(*) AS CNT
      FROM CHALLENGES
      GROUP BY HACKER_ID
    ) AS TEMP
    GROUP BY CNT
    HAVING COUNT(*) = 1
  )
ORDER BY
  TOTAL_CHALLENGES DESC,
  H.HACKER_ID;
```

Question 2:

You did such a great job helping Julia with her last coding contest challenge that she wants you to work on this one, too!

The total score of a hacker is the sum of their maximum scores for all of the challenges. Write a query to print the hacker_id, name, and total score of the hackers ordered by the descending score. If more than one hacker achieved the same total score, then sort the result by ascending hacker_id. Exclude all hackers with a total score of 0 from your result.

Input Format

The following tables contain contest data:

Column	Type
hacker_id	Integer
name	String

- Hackers: The hacker_id is the id of the hacker, and name is the name of the hacker.
- Submissions: The submission_id is the id of the submission, hacker_id is the id of the hacker who made the submission, challenge_id is the id of the challenge for which the submission belongs to, and score is the score of the submission.

Column	Type
submission_id	Integer
hacker_id	Integer
challenge_id	Integer
score	Integer

Solution:

```
SELECT
   H.HACKER_ID,
   H.NAME,
   SUM(X.MAX_SCORE) AS TOTAL_SCORE
FROM HACKERS AS H
JOIN (
   SELECT
      HACKER_ID,
      CHALLENGE_ID,
      MAX(SCORE) AS MAX_SCORE
   FROM SUBMISSIONS
   GROUP BY
      HACKER_ID,
      CHALLENGE_ID
) AS X
   ON H.HACKER_ID = X.HACKER_ID
GROUP BY
   H.HACKER_ID,
   H.NAME
HAVING
   SUM(X.MAX_SCORE) > 0
ORDER BY
   TOTAL_SCORE DESC,
   H.HACKER_ID;
```